Construction of a Unified Knowledge Graph for Cyber Threat Intelligence

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***Abstract*—The rapid expansion and variety of cyber-threat information put enormous pressure on security operations centers (SOCs) that must convert unstructured data into understandable signals and make decisions upon it. This paper develops a Cyber-Threat-Intelligence (CTI) framework that integrates vulnerability information, product inventories, and weakness taxonomies into a domain-specific knowledge graph via automatic fusing. The proposed solution covers 284,296 CVEs, 101,644 CPE identifiers, and 965 CWE weaknesses, generating more than 800,000 typed edges linking threats, assets, tactics, and mitigations in an integrated CTI Knowledge graph. The graph was cross validated against four external standard datasets achieves full coverage of ATT&CK CAPEC, STIX, and CVE-CAPEC mappings, 98.83 % node recall for CWE references, and edge-level full coverage for CAUSES_WEAKNESS and 99.83 % for AFFECTS; EXPLOITED_BY relations reach 74.58 % precision and recall. These results confirm that the integrated graph preserves structural integrity while substantially consolidating fragmented CTI sources, by consolidating fragmented CTI sources into a semantic model of high fidelity that could reduce analyst workload but also enables SOC workflows to be more efficient, thereby laying the groundwork for more proactive and intelligent cyber-defense capabilities.**

Keywords—cybersecurity, cyber threat intelligence, knowledge graphs, security operation automation

I. Introduction

Cyber Threat Intelligence has emerged as a crucial component of proactive cyber defense, referring to the collection and analysis of information about threats and threat actors to inform security decisions [1,2]. By providing context on adversaries' tactics, techniques, and procedures (TTPs), CTI enables organizations to anticipate and mitigate attacks more effectively [1,3]. Established models such as the Lockheed Martin *Kill Chain* framework [4] and the *Diamond Model* of intrusion analysis [5] laid early groundwork for understanding and classifying attacker behaviors. In parallel, standardized formats for sharing threat data – most notably the Structured Threat Information eXpression (STIX) [6,7] were developed to facilitate the exchange of threat intelligence among organizations. However, the overwhelming volume and variety of threat data being produced pose significant challenges for security teams [8-10]. Analysts must sift through massive amounts of heterogeneous data, such as Indicators of Compromise (IoC), logs, reports, etc., which is time-consuming and prone to oversight [8]. For example, a 2019 industry survey found that over half of security teams spent more than 40% of their time processing unstructured threat feeds and reports [10]. These issues highlight the need for approaches that can aggregate and analyze threat information efficiently.

The use of knowledge graphs has recently proved its effectiveness in representing complex information [11]. A knowledge graph is a semantic network of entities that are represented as nodes and relationships represented as edges that can fuse data from disparate sources into a unified, queryable form. Within cybersecurity, researchers have explored ontologies and knowledge graphs to capture relationships among threats, vulnerabilities, threat actors, and defensive measures. For instance, the Unified Cybersecurity Ontology (UCO) extended STIX with a formal ontology to integrate different threat information sources [12]. Similarly, Bridges *et al.*[13] built a cybersecurity knowledge graph by fusing malware incident reports, enabling analysts to query and reason about attacks in a structured manner [13]. By connecting indicators to higher level patterns, knowledge graphs promise greater situational awareness. Analysts of these knowledge graphs can discover hidden links (e.g. multiple IoCs relate to the same campaign) and draw on context that would be lost in isolated data feeds [12,13]. Recently advances in artificial intelligence, especially machine learning (ML) and deep learning, are being leveraged to automate threat analysis tasks [14,15]. For example, deep neural networks have improved intrusion detection and malware classification by learning complex patterns in data [14]. In the CTI domain, natural language processing (NLP) techniques powered by deep learning can extract IOC entities and attack techniques from unstructured text such as threat reports [16], reducing the manual burden on analysts. Nevertheless, current solutions meet challenges providing timely, actionable intelligence due to fragmented data silos and the dynamic, evolving nature of threats.

This paper is organized as follows: Section II provides a background on Knowledge Graphs, Graph Neural Networks, and their relevance to cybersecurity. Section III presents a detailed literature survey cyber threat intelligence field, focusing on their methodologies and results. Section IV discusses the proposed cyber threat intelligence framework phases. Section V outlines potential future research directions,

II. Background

Cyber threats refer to dangerous activities that have the intention of either breaking into the computer systems,

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networks, or data of others or disrupting these systems. The sources of such threats are nation-states, terroristic groups, criminals, individual hackers, and malicious insiders. They use various tools of damage such as malware, social engineering, supply chain attacks, man-in-the-middle attacks, and denial-of-service attacks [38].

Earlier, the typical way of dealing with cybersecurity issues has always been defensive after the breakout, a practice that can be very expensive. In fact, the worldwide loss from cyber-related criminal activities is estimated to be around $6 trillion by the year 2021 [39]. The reactive method mentioned above is not only responsible for financial and reputational costs but also for the lost system up time, incident responses, and reputation. So, the experts believe that the smart decision for a sustainable and safe world is to turn to a forward-looking approach. Cyber Threat Intelligence is a proposed solution. The approach engages in the collection, examination, and distribution of information about both the new and the existing forms of threats. CTI is an approach through which organizations are able to predict the expected cyber threats and defeat them using their security measures [39]. The following sections trace the path from the CTI lifecycle and sharing standards, through knowledge graphs that unify this intelligence, to deep-learning tools that automatically populate and interrogate the graph.

A. Cyber Threat Intelligence (CTI)

In general, CTI can be defined as the analysis of data-based knowledge about current and probable menaces to the main sectors of the institutions that can later be taken into account for security decisions [1,2]. These data include, usually, information about the intentions or objectives, the available resources, the tactics, the techniques, procedures, and the indicators of compromise that can be obtained from incidents of the past occurrence [1]. The CTI lifecycle consists of the stages of collecting raw data, then analyzing and converting it into information, and finally, transmitting intelligence to the interested parties [3]. In reality, companies utilize CTI derived from different sources such as technical feeds such as malware signatures, blacklists, human reports (threat actor profiles, incident reports), and open-source intelligence. To manage the sharing and consistency of the information, more than one standard and a number of frameworks have been given the nod of approval. As an example, the MITRE ATT&CK framework [17] is a repository of adversary ways and means structured in a way that it is popularly used to represent threats. CTI has often been the subject of sharing in formats such as the STIX [6] language, which is a standard set of tools that the community can understand. STIX has undergone various changes, the most recent of which, the 2.1 specification [7], has been more expressive and uniform, and it can be used along with the vehicle protocols such as TAXII to transfer the intel between the entities. Moreover, the MISP [21] is one of the group tools that has put into practice collaborative sharing of the threats warnings and the cases in real time. Nevertheless, people are still faced with difficulties when they try to share threat intelligence among different organizations due to trust and data sensitivity, and format inconsistencies. Standards creators and enforcement agencies have made an extra attempt to enable CTI sharing by issuing guidelines such as NIST's Special Publication 800-150 [22], which enumerates the most suitable methods and strategies for the execution of cyber threat information sharing partnerships [22]. Besides, the outcomes of the conducted surveys in such a sphere as the industry point to the continuous existence of the issues standing in the way of sharing, notably legal strictures and the failure to automate [23]. Enhancing the ability of intelligence to move across the interfaces and automating the transfer of threat intelligence are the main current research and development efforts [19, 20].

B. Knowledge Graphs

A knowledge graph is essentially a graph-based data model that links entities (nodes) with relationships (edges) under a defined ontology. One of the examples of entities in the cybersecurity context could be IP addresses, domain names, malware families, threat actor groups, vulnerabilities, tool. And it is the relation between these entities (e.g. “IP address X indicates malware Y” or “threat actor A uses tool B”) that relationships express. A cybersecurity knowledge graph (CKG) that exhibits such relationships is capable of offering a more comprehensive view of the threat landscape. Thus far, the fundamental methodologies and schemas essential for constructing these graphs have been studied by previous research.

Unified Cybersecurity Ontology [12], people building the field have been able to outline the layers of the domain. UCO, developed by Syed et al., defines the classes and their interlinkages, enabling the integration of data coming from various sources (e.g., Malware is a type of Indicator, that Campaigns are made of Incidents, etc.)

While the remaining studies concentrate on constructing the graphs from preexisting data sources: for example, the study of Bridges et al.[13] revealed that the knowledge graph of the cyber threat could be created with the information from the malware incidents reports via the proposed parser, and the graph could further be queried for finding the patterns. Similarly, the paper by Liu et al.[24] describes an ontology-based strategy aiming at the network security knowledge graph aligned to the STIX framework, which represents the conversion process of standard threat reports to a graph.

Cybersecurity knowledge graph provides visualization that support various tasks like threat hunting that is the process to search for, identify, and remove malware from the enterprise network or for anomaly detection to discover the presence of abnormal traffic. For example, one might not notice it, A well-structured CKG supports capabilities like threat hunting and correlation: by traversing the graph, an analyst might find that seemingly unrelated indicators are connected via the same threat actor or attack group, improving situational awareness. Notably, the ATT&CK framework has been used as a spine for organizing such knowledge. Recent work by Bolton et al.[18] mapped several existing cybersecurity knowledge graphs to the MITRE ATT&CK tactics and techniques matrix, revealing coverage gaps and helping ensure that a CTI program addresses the full spectrum of adversary behavior. Overall, knowledge graphs and semantic web tools introduce a level of organization and reasoning to CTI that goes beyond simple indicator feeds. Finally, the graph model introduced into the CTI sector is an organizational tool for ontologies. The other part of the stage is the language that the model uses to communicate. To expound, as the graph model was incorporated to the CTI area, clear power is restored, and the

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manager of the system can now decide what to do in the process at highs and lows.

C. Deep learning and Automation

Given the scale and complexity of threat data, automation is critical. Machine learning, and in particular deep learning, has been increasingly applied in cybersecurity to automate detection and analysis tasks [14][15]. In the realm of CTI, one important application of ML is natural language processing for threat intelligence extraction. A considerable portion of CTI arrives as unstructured text (incident reports, threat advisories, blog posts). NLP techniques can convert this text into structured intelligence by identifying mentions of threat entities and the relationships between them. For example, Husari *et al*[16]. developed TTPDrill, which automatically extracts attacker actions (TTPs) from unstructured CTI reports using a combination of parsing and machine learning. In recent years, researchers have trained deep learning models (e.g. using LSTM and BERT architectures) to perform joint entity and relation extraction in cybersecurity documents [30][31]. This line of work treats CTI text as a sequence and learns to tag indicators (IP, file hash, vulnerability, etc.) and link them to the corresponding attack techniques or threat actors mentioned. Automating this step is essential for building knowledge graphs at scale. Another growing area is the use of graph-based machine learning on the knowledge graph itself. Graph neural networks (GNNs) can operate on the structure of a knowledge graph to perform tasks like node classification, link prediction, or anomaly detection. Wang *et al.*[25], introduces a relational graph attention network for learning embeddings of CTI knowledge graph entities – such embeddings can help in discovering new connections (e.g. predicting that a new IOC is associated with a known campaign) or verifying the consistency of the intel graph. There is also emerging interest in applying large language models (LLMs) to CTI problems. Early studies have shown that generative LLMs like GPT-3 or ChatGPT can assist in processing threat reports and even suggest structured intelligence outputs [34]. Tang *et al.* [34] demonstrate an approach where ChatGPT is leveraged to extract cyber threat indicators and populate a knowledge graph, illustrating the potential of LLMs to augment CTI workflows. While still nascent, such AI-driven automation could significantly reduce the manual burden on analysts and improve the timeliness of threat intelligence.

Additionally, the KG-LLM framework [13] illustrates the benefits of combining LLMs with knowledge graphs, significantly improving link prediction capabilities and generalization to new or unseen threats. These advancements demonstrate the powerful potential of integrating LLMs with knowledge graphs to streamline threat intelligence extraction and reasoning processes.

III. LITERATURE SURVEY

Researchers have studied the application of knowledge graphs, semantic technologies, and machine learning to enhance cyber threat intelligence. A number of frameworks for constructing threat intelligence knowledge graphs from unstructured or semi-structured data have been proposed. One prominent research is the work by Piplai *et al.* [26], who created a cybersecurity knowledge graph by extracting information from after-action malware reports. Their system, built on an extension of STIX, was able to integrate malware indicators, behaviors, and relationships into a queryable graph, facilitating incident response and threat hunting. Another notable system is Open-CyKG by Sarhan and Spruit[27], an open CTI knowledge graph that automates the extraction of threat information focusing on advanced persistent threat groups using an attention-based neural Open Information Extraction model. Open-CyKG demonstrated that fusing data from raw threat reports into a knowledge graph can outperform manual methods in completeness and speed. Li *et al.*[28] introduced AttacKG, a framework to automatically construct a knowledge graph of attack techniques by parsing CTI reports. AttacKG focuses on mapping detailed attacker techniques aligned with ATT&CK from text, effectively bridging unstructured CTI with a structured technique-centric graph. These efforts show a clear evolution, earlier works established[26,27], that it is feasible to generate CTI knowledge graphs automatically, and more recent works [18,28] have homed in on aligning the extracted knowledge with standardized schemas and adversary models e.g. ATT&CK.

Extracting entities and relationships for knowledge graphs in the field of cyber intelligence has recently been a hot research topic. researchers have increasingly applied deep learning-based NLP pipelines. Early methods, such as the Extractor system by Satvat *et al.* [29], used deep neural networks to identify attack behaviors described in threat reports, achieving notable accuracy in parsing sentences into triples (e.g. *malware, uses, technique)*. Recent approaches have made further enhancements where Mouiche and Saad [30] proposed a threat intelligence knowledge graph pipeline that employed bidirectional LSTMs and attention mechanisms to perform named entity recognition and relation extraction on cyber incident texts. Their pipeline was evaluated using security reports, and was able to automatically populate a knowledge graph with extracted entities such as exploit targets, malware names and the relations between them. Similarly, Ahmed *et al.*[31] developed CyberEntRel, a joint extraction model using transformer-based embeddings to simultaneously tag entities and identify their relations. By incorporating cybersecurity domain knowledge into the training process, their model achieved remarkable F1 in extracting threat entities and over 80% in extracting relations [31] – a substantial improvement that underscores the effectiveness of tailored deep learning models for CTI text. These automated extraction techniques are crucial: they form the front-end of any system that builds or updates a CTI knowledge graph from raw data.

Beyond the construction of knowledge graphs, researchers have examined how to analyze and utilize these knowledge graphs for threat intelligence. Spyropoulos *et al.* [32] describe an AI-based holistic CTI management framework that integrates a knowledge graph with machine learning analytics. Their framework developed within the EU-funded *SAPPAN* project. It aggregates threat data into a graph and then applies techniques like graph-based clustering and alert ranking to assist analysts; it demonstrates the value of combining a CTI knowledge graph with AI to prioritize

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relevant threats. In the realm of domain-specific applications, Al-Qirim *et al.* [33] presents a CTI system for smart grid SCADA environments that uses knowledge graphs alongside digital twins and machine learning. This approach allowed power system operators to map cyber-threat intelligence directly onto operational technology networks, yielding improved detection of attack scenarios targeting critical infrastructure [33]. The literature also shows a nascent interest in harnessing large language model**s** for CTI. Tang *et al.* [34] illustrate how an LLM (ChatGPT in their case) can be used to parse raw threat intelligence text and help construct a knowledge graph by suggesting entities and relationships. While LLMs are not yet widely adopted in deployed CTI systems, this exploratory work suggests they could become a valuable tool for extracting actionable intelligence from the vast amounts of unstructured data.

Finally, considerable work has been done on sharing and maintaining threat intelligence knowledge. Bromander *et al.* [35] investigated the challenges of sharing CTI between organizations and proposed a new data model and ontology to automate CTI exchange. Their model emphasized structured knowledge representation using graph-based schemas to enable tool interoperability and reduce the manual effort in translating threat intel from one format to another [35]. In a similar vein, Ammi *et al.* [36] leveraged a cloud-native architecture to semantically interconnect threat data from different sources, effectively creating an ecosystem where a shared knowledge graph is continuously enriched by multiple contributors. This shows that beyond building isolated knowledge graphs, the community is also focusing on collaborative and scalable CTI knowledge networks. Furthermore, efforts like Huang *et al.* [37] have combined ontologies with established frameworks (mapping a cybersecurity ontology to the MITRE ATT&CK tactics and techniques) to support automated reasoning about adversary behavior. By aligning knowledge graph content with widely used models of attacker behavior, such approaches can assist in attributing incidents to threat groups or predicting an adversary's next moves based on their TTP profile [18,37].

Recent work in cyber-threat intelligence is in a process of change from isolated databases and one-off text-mining exercises to integrated knowledge-graph platforms which merge various sources vulnerability feeds, adversary tactics, asset inventories, and sensor telemetry into a single reasoning space. The first ontology efforts paved the semantic way for the integration of these kinds of works [12,19], while the next sharing frameworks gave the example of how multiple organizations can populate one graph without losing the provenance of the data [20,35]. Besides, the automating construction of graphs from raw reports and feeds [30,31] has been implemented by researchers and the next step of the process is to perform graph-centric analytics such as link prediction, neighborhood scoring, motif discovery to forecast emerging attack paths or to emphasize those that were hidden before among weaknesses, exploits, and defensive controls [32,33]. The areas of these studies together demonstrate the predictive power and the explanatory value coming from the process of disparate data normalizing, inter-linking, and traversing inside the graph. Thus, our work intends to propose a strategy that enriches the single CTI knowledge graph in near-real time, intending to provide faster and more context-rich information to analysts and decision-makers.

IV. Proposed Framework

Organizations encounter more sophisticated cybersecurity challenges that call for proactive and effective interventions. To solve the issue, the proposed framework integrates heterogeneous sources of information aimed at identifying, analyzing, and responding to cyber threats in real time, improving organizational security. As illustrated in Fig 1 below, the proposed framework is organized into five phases for building an intelligent KG-based CTI reliable, dynamic, and scalable prediction system. The first phase is the data collection phase where heterogeneous structured, semi-structured, and unstructured data are collected covering cyber threats concerning both cyber threats and computer assets, gathering vital information from numerous sources. Phase 2 is the construction of a domain-specific knowledge graph covering cyber threats and computer assets, encoding the relations and vulnerabilities contained within the data that has been collected. Phase 3 is Knowledge graphs integration where entity matching, and alignment techniques are applied. The output of this phase is an integrated domain-specific Knowledge graph representing concepts and interrelated relationships. Phase 4 adopts GNN algorithms for enhancing the integrated Knowledge graph by predicting links, thereby detecting possible vulnerabilities and attack trajectories. Phase 5 is focusing on applying techniques for generating insights such as Neo4j are employed taking the enhanced Knowledge graph as a knowledge base for monitoring and visualizing an effective and proactive cybersecurity system that could predict and counter impending threats.

A. Phase 1: Data Collection

This phase involves the process of extracting information from diverse sources To represent different concepts covering CTI such as cyber threats and computer assets. The collected data is used not only as a source for the subsequent phases but also as the basis for the construction of detailed and accurate knowledge graphs, which could be further used for analysis and threat detection. The data collection process is a two-step procedure: First, Cyber Threats data sources are collected from various threat intelligence feeds, security logs, and open-source reliable datasets such as MITRE ATT&CK. Thus, the IOCs, threat actor profiles, and historical attack data can be the main cyber threat intelligence. The second step is Computer Assets data sources, where data is retrieved from computer assets such as internal databases, and IT asset management tools that are the sources used for hardware, software and network data, for example OCS Inventory or Common Platform Enumeration (CPE). This comprises information on software vulnerabilities and affected versions.

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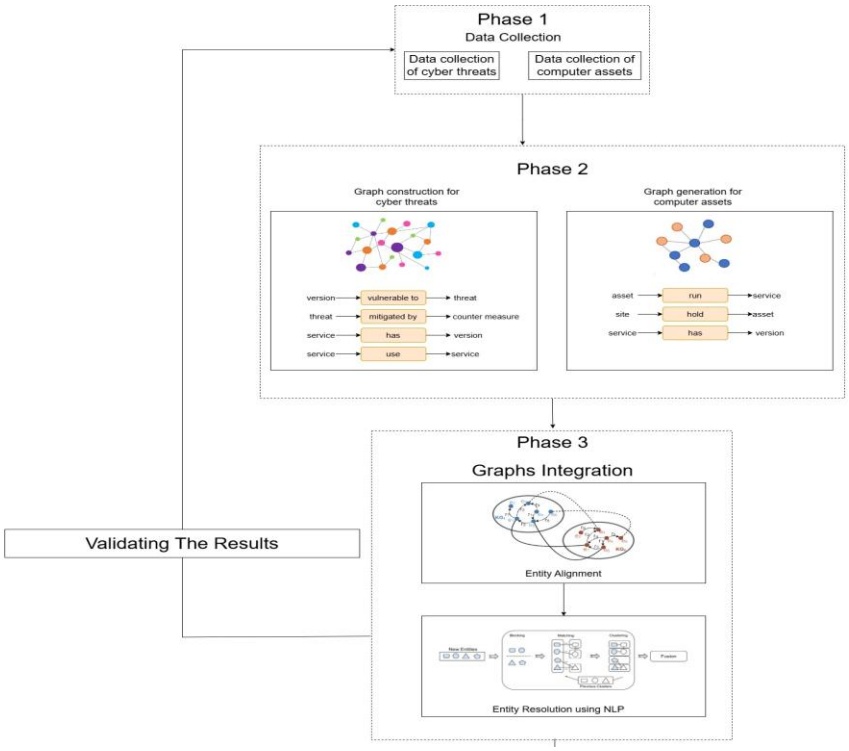


Fig 1. Cybersecurity Threat Intelligence Framework

B. *Phase 2: Graph Construction*

Graph Construction phase focuses on the construction of two main subgraphs cyber threat Graph Construction and Computer Asset Graph Construction.

1) Cyber Threat Graph Construction

The threat-centric layer of our knowledge graph is fashioned by nine node types that are connected by eleven edge categories. The central Weakness vertex, represented by 965 CWE instances, links to real-world evidence through 172,278 EXPLOITED_BY edges to CVE nodes. Tactical context is supplied by 1,212 CAUSE_WEAKNESS links to CAPEC AttackPattern nodes. Each pattern maps into the MITRE ATT&CK corpus via 2,852 USES_TECHNIQUE edges that terminate at 656 Technique vertices and aggregate to 14 Tactic vertices through IMPLEMENTS. Adversary attribution is captured by 159 Group nodes connected to the same techniques with USES_TECHNIQUE. Defensive knowledge is attached directly to the weakness by 1,525 MITIGATED_BY edges leading to 991 Mitigation nodes and 718 DETECTED_BY edges pointing to 19 DetectionMethod nodes. Platform scope is defined through 1,502 AFFECTS edges that link to 54 Platform nodes, while 1,683 IS_PARENT and 453 SUBTECHNIQUE_OF edges preserve the hierarchical structures of CWE and ATT&CK. This configuration enables analysts to traverse seamlessly from an attacker group's tactic to concrete software versions and the corresponding countermeasures.

2) Computer Asset Graph Construction

The asset-centric layer consolidates product and deployment information around three core concepts: CPE, Weakness, and Platform. The graph contains 101,644 CPE identifiers; each joined to its latent vulnerability by VULNERABLE_TO edges and thus inheriting the full CVE hierarchy. When version detail is available an AFFECTS_VERSION which is a sub-property of VULNERABLE_TO relationship distinguishes vulnerable from patched builds. Operational exposure is expressed by AFFECTS edges that link weaknesses to 54 platform categories such as cloud, mobile and SCADA.

The output graphs from this phase is used in phase 3. Phase 3 aims to integrate the Cyber Threat Graph and Computer Asset Graph into a single knowledge graph. This integration leads to the development of a domain-specific knowledge base necessary for representing cybersecurity concepts including external threat information and internal asset information

C. *Phase 3: Graphs Integration*

In this phase, an NLP pipeline is adopted involving two steps Entity Alignment and Entity Resolution. Similar entities existing in the two graphs are identified and aligned. This is imperative to avoid data redundancy and guarantee that all relevant concepts are represented correctly reflecting the real-world entities. For example, an IP address labeled as malicious in the Cyber Threat Graph should match its matching entity in the Computer Asset Graph, which may be a server or network node owned by the organization.

Fig 2 shows the entity alignment process, where the elements of the internal network graph (G), like "Server a" (a2), "Malware" (a1), and "Vulnerability" (a3), are aligned with the elements of the external threat intelligence graph (G'), like "Server b" (b2), "Threat Actor" (b1), and "Phishing Email" (b3). The alignment procedure facilitates a Product Combination Graph (PCG), in which alignments between aligned nodes (e.g., (a1, b1) and (a3, b3)) assist in revealing key threat relationships. To enhance the integration further, Entity Linking and Resolution techniques are applied. This is a crucial step to disambiguate and resolve conflicting entity references between datasets. Methods like Named Entity Recognition (NER) and word embeddings analyze the contextual setting, attributes, and relationships of entities to maintain consistency [40].

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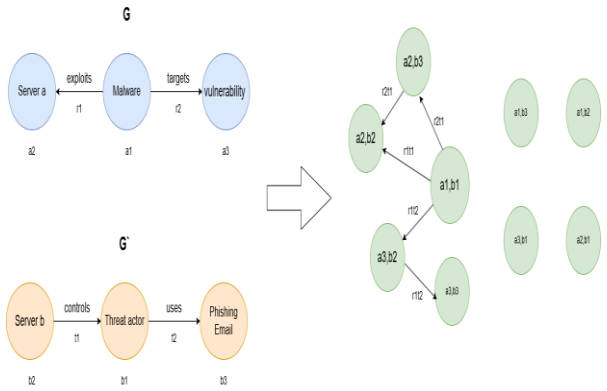


Fig 2. Aligning Entities of graph G and G'

Fig 3 demonstrates how various references to a single cybersecurity term—e.g., "Man in the Middle Attack" with its abbreviation "MITM"—are properly recognized as the same entity. Such an operation maintains coherence in the knowledge graph and avoids fragmentation in data representation [40]. After successfully achieving entity alignment and resolution, the next step is to merge aligned entities and related relations into an integrated structured knowledge graph. The integrated graph represents a scalable and dynamic representation of threats and assets is evident, with newly merging attack pathways, vulnerable assets, and mitigation methods surfacing. For instance, a Threat Actor (b1) that launches Malware (a1) to take advantage of internal vulnerabilities (a3) is presented as a defined relationship within the graph as shown in Fig 3.

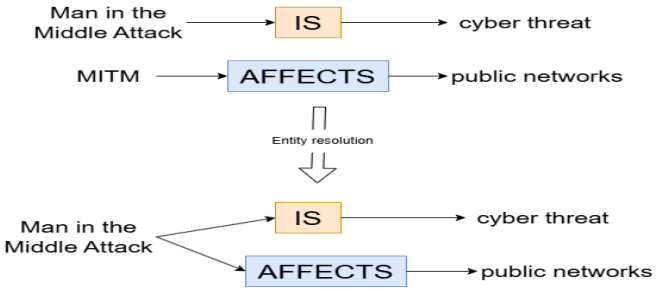


Fig 3. Entity Resolution

These findings enable cybersecurity professionals to predict and avert threats better by knowing how external parties communicate with internal systems.

The last step in this phase entails Validation and Evaluation techniques to verify the integrity and quality of the integrated knowledge graph. The evaluation of the graph is performed by aligning extracted entities and relations with domain-specific cybersecurity datasets [6,7,17,22]. When inconsistencies or data gaps are found, the integration process is refined iteratively until the graph is in a satisfactory cybersecurity domain-specific representation through comparison with existing cybersecurity knowledge bases. The result of this phase is a cybersecurity knowledge graph that integrates internal asset information with external threat intelligence, thus delivering an enhanced method of threat detection, analysis, and proactive defense. Using structured relationships and contextualized threat intelligence, organizations can anticipate cyber threats, determine vulnerabilities, and reinforce security posture. By consolidating, linking, and integrating these two graphs, organizations can improve their threat analysis, identify potential attack vectors, and improve their cybersecurity defense systems. There are several steps involved in the process to guarantee the accuracy, consistency, and comprehensiveness of the composite graph

In this study, we were engaged in the development of a full-fledged cybersecurity ontology by jointly combining the data obtained from the Common Weakness Enumeration (CWE) and the Common Attack Pattern Enumeration and Classification (CAPEC) databases. The chief objective of our research was to forge a systematic semantic representation, which is the basis for a more thorough investigation into the sources of cybersecurity threats and vulnerabilities

First, separate picture graphs were created which were related to the CWE and CAPEC datasets. Firstly, CWE delivered precise information related to weaknesses in the software, while CAPEC contained detailed information regarding the attack techniques' operation. One graph was made up of specific nodes that represented entities such as weaknesses, attack patterns, mitigation techniques, and detection methods with connections through relationships clearly indicating interactions like "causing weakness," "mitigated by," and "detected by".

To carry out the integration of these separate graphs, we applied string-matching methods on entity names as well as their descriptions. This style utilized text similarity to compare the two data labels and their descriptions so that they are accurately aligned and integrated.

Fig [4] illustrates the final knowledge graph structure after the integration, highlighting the main entities and their connections. The final unified graph shows an enhanced ability of cyber threat intelligence (CTI) and the provision of cybersecurity professionals and organizations with the useful tool to recognize threats and mitigate them proactively.

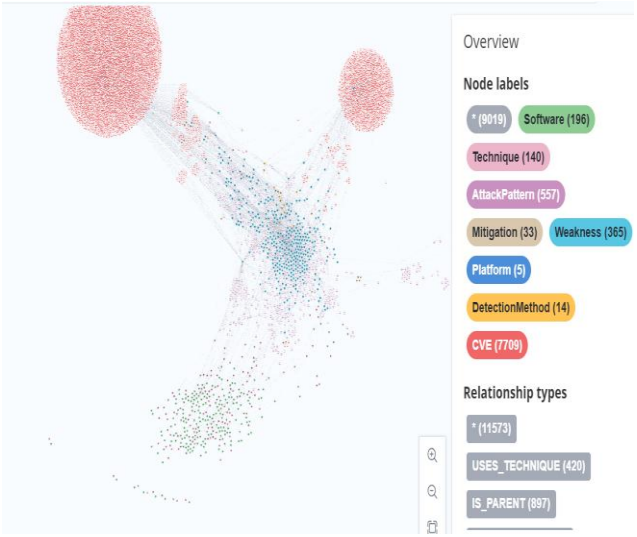


Fig 4. Graph construction sample

We evaluated the constructed knowledge graph across a series of validation tasks covering both node coverage and edge-level accuracy [41]. Table 1 summarizes the key metrics coverage (for node level comparisons) and precision, recall, and F1 score (for both node and edge level tasks).

Table 1. Experiment Results

Task	Ref	KG	Hits	Rec (%)	Prec (%)	F1 (%)
CAPEC: ATT&CK slice	17	17	17	Fully covered	–	–

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CAPEC: STIX bundles	32	32	32	Fully covered	–	–
CAPEC: NVD CVE→CAPEC (20–24)	20	20	20	Fully covered	–	–
CWE: NVD feeds (20–24)	341	965	337	98.83	37.31	54.14
CAUSES_WEAKNESS (edges)	1212	1212	1212	Fully covered	–	–
EXPLOITED_BY (edges)	354	354	264	74.58	74.58	74.58
AFFECTS (edges)	1507	1502	1502	99.67	100.0	99.83

Node-level evaluation. CAPEC nodes drew on three independent mappings—MITRE ATT&CK crosswalk, STIX bundles, and the NVD CVE→CAPEC cross-walk (2020–2024)—and in every case the graph contained 100 % of the expected IDs (17/17, 32/32, and 20/20, respectively). For CWEs, we compared against the NVD feeds from 2020 to 2024: 337 of the 341 referenced CWE IDs appeared in the graph (98.83 % recall). Framed as a retrieval task, this yielded 98.63 % recall, 37.31 % precision, and a 54.14 % F_1, reflecting additional KG-unique CWE nodes beyond those cited in NVD.

Relationship-level evaluation. Three different gold standards were used. CAUSES_WEAKNESS edges were validated against the BRON JSON database, matching all 1212 edges for perfect precision, recall, and F1 (100 %). AFFECTS edges were benchmarked on platform mappings extracted from NVD CPE feeds (2002–2024), confirming 1502 of 1507 edges (99.67 % recall, 100 % precision, 99.83 % F_1). EXPLOITED_BY links were first tested against the NVD-derived CWE→CVE list, and the 90 remaining unvalidated links were then cross-referenced using the CVE2CAPEC dataset; overall, 264 of 354 edges validated (74.58 % precision, recall, and F1), making this the primary area for targeted enrichment.

Fig 5 illustrates a two-hop sub-graph centered on CWE-732 (Incorrect Permission Assignment for Critical Resource). The visualization reveals a complete attack chain that traces the weakness to real-world exploits and corresponding defensive controls. Edges of type EXPLOITED_BY connect CWE-732 to two publicly disclosed vulnerabilities CVE-2019-25091 and CVE-2021-34563 demonstrating its practical impact across disparate products. Upstream, the weakness is causally linked to the CAPEC-1 attack pattern (*Accessing Functionality Not Properly Constrained by ACLs*) and to the MITRE ATT&CK technique T1574.010 (*Services File Permissions Weakness*), thereby situating the flaw within both tactical and strategic threat-model frameworks. Conversely, MITIGATED_BY and DETECTED_BY relationships enumerate concrete remediation and detection artefacts ranging from automated static analysis to least-privilege configuration guidelines. At the same time, AFFECTS edges mark the weakness as platform-agnostic but especially relevant to web-based deployments. Finally, hierarchical IS_PARENT links expose semantically related CWEs, enabling propagation of countermeasures to broader or more specific vulnerability classes. Collectively, this sub-graph offers a narrative that maps a weakness to exploitable instances, attacker methodologies, recommended mitigations, and environmental scope, illustrating how knowledge-graph analytics can transform raw CTI data into actionable, context-rich intelligence for both researchers and practitioners.

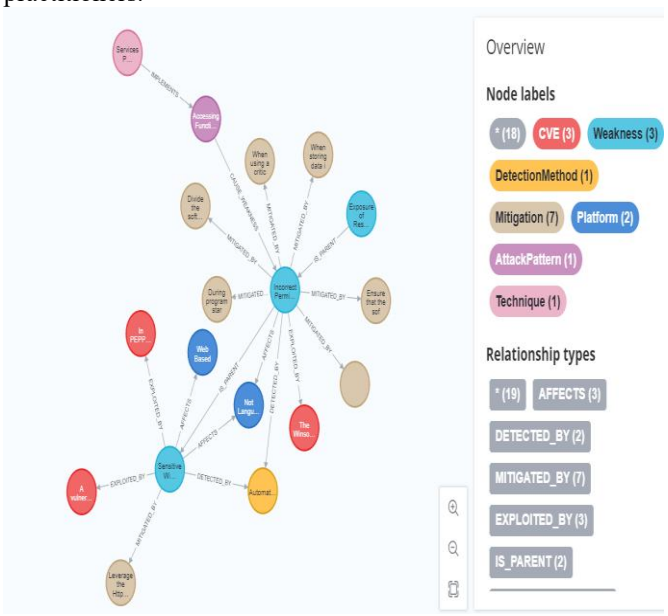


Fig 5. Sub-graph visualization

V. Conclusion

This paper presented a framework for constructing a unified cyber threat intelligence knowledge graph that integrates heterogeneous cyber threat and asset data into a single, domain-specific knowledge graph. This knowledge graph captures complex relationships among threats, vulnerabilities, and organizational assets, creating a semantic model of the cyber landscape. Building on this foundation, graph neural networks could be applied for link prediction and anomaly detection, enabling the discovery of hidden threat-asset links and the identification of abnormal patterns indicative of emerging attacks. Through these graph-based deep learning analytics, the system is expected to uncover unseen attack paths and identify possible suspicious activities within the network, thereby significantly enhancing proactive threat detection.

The constructed knowledge graph was validated using four external standard datasets, which represent tests for structural integrity and semantic accuracy. The graph achieved in coverage of ATT&CK–CAPEC crosswalks by 100%, validation for CAUSES_WEAKNESS relations by 100%, and AFFECTS edges accuracy of 99.8%. Moreover, 98.83% of print CWE nodes were captured from NVD feeds and the system achieved an F1-score of 54.14% in node-level retrieval thus reflecting the graph's enrichment beyond standard datasets. These findings demonstrate the framework's capacity to integrate threat intelligence and the data for internal assets into an understandable, query compatible format.

VI. Future Work

There is still a need to improve result quality both in terms of precision and recall when inferring relationships such as EXPLOITED_BY. Higher-fidelity entity linking, more

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balanced training sets, and confidence-aware ensemble models could reduce false positives while uncovering low-frequency threats that currently elude extraction, another major challenge is scalability and performance. Exploratory queries over approximately one million edges already place heavy demands on conventional graph stores; investigating distributed property-graph backends and dynamic embedding schemes that update vertex representations without full retraining may enable sub-second link prediction on billion-scale graphs. Furthermore deep-learning extractors and graph neural networks occasionally hallucinate or propagate spurious relationships [43], which risks analyst distrust. Embedding provenance metadata, uncertainty scores, and counterfactual path explanations directly into query results alongside adversarial-robust training techniques could enhance transparency and resilience.

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